MIL-T-22536A(AS)
2 January 1969
Superseding
MIL-T-22536(Wep)
20 June 1960

MILITARY SPECIFICATION
TRACTOR, AIRCRAFT, TOWING, LOW PROFILE SPECIAL DUTY, 8500 LBS. DRAWBAR PULL TYPE MD-3

This specification has been approved by the Naval Air System Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the manufacture and operating requirements of a diesel powered tractor intended for towing all carrier based aircraft aboard aircraft carriers. This tractor includes an electrical power receptacle for use with aircraft. This tractor shall be designed to mate with a turbine carrier housing and auxiliary power unit (APU). The APU is used for starting conventional and impingement type aircraft. The turbine carrier may be installed on the rear of the tractor. A rear counterweight is to be used in lieu of the trubine carrier.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and other applicable publications in effect on date of invitation for bids, form a part of this specification. Where there is a conflict between this specification and the referenced specifications, standards, or other publications, this specification shall govern:

SPECIFICATIONS:

Federal

ZZ-T-381	Tires, pneumatic; Vehicular (Highway)
ZZ-I-550	Inner Tubes; Pneumatic Tire

FSC 2420

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the currentversion before use.

SPECIFICATIONS: (Continued)

Military

MIL-P-514	Plates, Identification, Instruction, and Marking; Blank
MIL-S-901	Shock Tests, H. I. (High Impact) Shipboard Machinery
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-B-8565	Batteries, Storage, Aeronautical
MIL-V-13811	Varnish, Waterproofing, Electrical, Ignition

STANDARDS:

Federal

FED-STD No. 595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristic Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground Support Equipment
MIL-STD-831	Test Reports, Preparation of
MS 18045	Batteries, Storage, Aircraft

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

STANDARDS: (Continued)

Military

MS 24523	Switches, Toggle
MS 3349(AS)	Connector, Plug, Electric Quick Disconnect, Battery
MS 51074	Horn, Electrical Automotive Type 24-Volt D.C.
MS 51302	Stoplight, Vehicular - Blackout
MS 51329	Stoplight - Taillight, Vehicular
MS 51330	Stoplight - Taillight, Vehicular

Aeronautical

AN 2552	Receptacle - External Power

Government Drawings:

ORD 7358618	Dash Light

Publications:

(When requesting specifications, standards, and publications refer to both the title and number. Copies of this specification may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Road, Philadelphia, Pennsylvania, 19120.)

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General: The tractor covered by this specification shall be complete and ready for operation when delivered. Tractor shall have a short, compact wheel base suitable for maneuvering in restricted areas and shall be constructed to meet all performance requirements and standards when tested as specified herein. The tractor shall be a product of a manufacturer engaged in the fabrication and production furnishing essentially similar catalog items for a period of two years prior to date of invitation for bids.

3.2 Preproduction Sample: Regarding the operating requirements not specifically covered by this specification, the tractor shall be equal in performance to the preproduction sample tractor. The design, structural, and operational characteristics of this sample will be evaluated for acceptability from a standpoint of safe and efficient operation and for conformance with generally accepted design practice and with provisions of this specification. The first tractor produced under the terms of a contract, following notice of award, shall be submitted for approval as a preproduction sample and approval shall be obtained prior to manufacture of additional tractors. Manufacture of additional equipment prior to approval of the preproduction sample will be at the contractor's risk and such additional items are subject to rejection if they fail to meet the requirements of this specification. The preproduction sample tractor, when approved, will be used as a standard for all tractors subsequently manufactured for a given contract under this specification and will be presented for inspection as the last unit for delivery under the contract. Examination and testing, wherever practical, will be conducted at the manufacturer's plant by a representative of the Contracting Officer, and the delivery schedule, under the terms of the contract, will automatically be extended in an amount equal to, but not exceeding, the time required for such testing and examination. The estimated time for testing is less than 10 days.

3.3 Material: Materials shall be as specified herein. Material not definitely specified may be manufacturer's normal commercial product, but shall adequately meet the test requirements specified in Section 4. When the term "normal commercial product" is used, it shall mean the item produced by a manufacturer normally engaged in commercial production of the item(s) covered by this specification. The Government reserves the right to require any bidder who proposes to furnish other than a "normal commercial product" to submit, prior to and as a condition preceding the award of any contract, detailed engineering drawings of the item(s) he proposes to furnish for the approval of the Contracting Officer.

3.3.1 Structural Design: The structural and mechanical members and load carrying parts shall be designed and constructed to withstand the stresses

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

resulting from shock loads and other conditions present in severe field operation without causing looseness, permanent deformation, or undue wear of any member or part.

3.4 Propulsion

3.4.1 Power Train Components: The power train shall consist of the following components: an engine, torque converter, automatic transmission, driveshaft, and axles. All components of the power train shall be compatible and capable of transmitting maximum engine torque through intervening gear reductions.

3.4.2 Engine: The diesel engine shall have the following features:

a. A six cylinder, two or four stroke cycle, liquid cooled engine

b. Have a piston displacement of at least 280 cubic inches

c. Develop not less than 100 HP at a governed speed

d. Maximum peak torque shall not be less than 200 foot-pounds at approximately 2000 rpm

e. The engine shall be certified for operation at a maximum speed of 3000 rpm

f. The governor shall be set and sealed at 3000 rpm

g. Engine idling speed shall not exceed 600 rpm

h. The engine shall perform efficiently using diesel or JP-5 fuel

3.5 Winterization: The tractor shall be designed for full operational use in any ambient temperature between -25°F and +125°F without the use of external heaters or special winteriation equipment.

3.6 Electrical System

3.6.1 Starting, Lighting, and Regulating Devices: The tractor shall be provided with a complete electric lighting system, amply equipped with safety and regulating devices. All control switches shall be within convenient reach of the driver. Ground polarity of the system shall be negative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

3.6.2 Power for Servicing Aircraft: In addition to providing electrical power for the tractor, the electrical system shall provide electrical power outlets for servicing the aircraft through a receptacle conforming to standard AN 2552. The electrical system shall be a 28-volt system designed and fabricated in accordance with MIL-STD-704. The electrical system shall be capable of withstanding the loading as follows:

a. Transient inrush of 400-500 amperes to the hydraulic pump motor, approximately 1/20 second duration

b. 80 amperes for 10 seconds

c. 35 amperes continuously, (motor idling on line prior to demand on hydraulic system)

d. 80 amperes for 5 seconds with 2 seconds rest, repeated for 5 cycles

e. 35 amperes continuously, (until power plug is removed from aircraft)

3.6.2.1 Electrical Power Receptacle: The electrical power receptacle for supplying electrical power to the aircraft described (in paragraph 3.6.2) shall be located on the left side of the rear of the tractor for convenient electrical hook-up to aircraft.

3.6.3 Alternator: The capacity of the alternator shall be 28-volts, dual rated at 60/40 ampere. It shall be totally enclosed and waterproofed to assure full performance in all environmental operations.

3.6.4 Starting System: The starting system shall consist of: starting motor, solenoid relay switch, neutral safety switch, starter button, and the necessary wiring and heavy duty cables properly assembled in a complete system. The starter button shall be located on the instrument panel where it can be conveniently operated by hand.

3.6.5 Starter: The starter motor shall be 24-volt, of adequate capacity, securely mounted, and equipped with a pinion-meshing device or over-running clutch. The starting motor shall be effectively protected from exhaust heat.

3.6.6 Charging Receptacle: A receptacle, conforming to Standard AN2552, and located on the right side of the tractor, in the forward portion, shall be provided for connection for emergency power from an external power source. Proper size load resistor(s) shall be placed between the receptacle and the

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.

starting motor to prevent excessive current from damaging the starting motor when tractor is being started by any 28-volt APU unit. The receptacle shall be protected by a spring loaded cover to prevent entrance of snow and water when not being used.

3.6.7 Neutral Safety Switch: A neutral safety switch wired in series with the starting circuit shall be connected to the transmission linkage so that the starting circuit will open and remain open while the transmission linkage is moved out of the neutral position.

3.6.8 Starting Motor Switch: The starting switch and circuit shall be designed to prevent starter engagement with the ring gear while engine is operating.

3.7 Lighting System, Battery, and Horn

3.7.1 Headlights: The tractor shall be equipped with two headlights of at least 50 watts each. The lights shall be sealed beam type, Trade No. 4593, and shall be installed above and behind the front bumper and recessed to prevent damage. Lights shall be horizontally adjustable.

3.7.2 Marker Lights: The tractor shall be equipped with two front marker lights, type MS-51302-21, recessed or adequately guarded to protect them against damage.

3.7.3 Headlight/Marker Lights Off - On Switch: The Off - on switch, type MS-24523-21, shall be mounted on the instrument panel, and shall control the headlights and front marker lights.

3.7.4 Stop and Taillights: Two combination stop and taillights shall be mounted on the rear of the tractor, and shall be adequately recessed or guarded to protect against easy damage. The stop light shall operate automatically upon depression of the service brake. The stop and taillights shall conform to MS-51329-1 and MS-51330-1, and shall be mounted on the left rear and right rear of the tractor respectively. Both the service tail and stop lights and the blackout tail and stop lights shall be controlled by one switch, MS-24523-23.

3.7.5 Pintle Light: One white light of not less than 15-candle power with a white lens cover shall be mounted on the rear of the tractor for illuminating the rear pintle and tow bar. This light shall be adequately recessed or guarded to protect it against damage. The pintle light on-off control switch shall be mounted on the instrument panel.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.

3.7.6 Storage Batteries: The tractor is to be equipped with two 24-volt storage batteries, type MS-18045-42, Federal Stock Number RM6140-782-7204 D336. Batteries shall be in accordance to MIL-B-8565, and shall be equipped with quick disconnect plugs, type MS-3349(AS).

3.7.7 Storage Battery Mountings: The batteries shall be in an easily accessible compartment and readily removable. If mounted outside, the batteries shall be housed in an individual box, which shall be fitted with a quick release cover to provide for easy inspection and servicing. Covering and positioning shall adequately protect against short circuit contact of electrical wiring and battery straps.

3.7.8 Horn: Tractor shall be equipped with an electric horn, MS-51074-1. The horn shall be operated by a push button located on the instrument panel. The horn button and wiring for the horn shall be adequately protected against moisture and inclement weather conditions. A metal plate reading "HORN" shall identify the horn button.

3.7.9 Electromagnetic Interference Limits: The tractor shall not emit electromagnetic interference radiation in excess of the limits established for REO 5, Class 111c in MIL-STD-461.

3.8 Fuel System: The fuel system shall consist of: fuel tank, fuel filters, fuel injection pump, connecting fuel lines, and an electric, thermally stabilized, remote indicating fuel gauge. The fuel system shall conform to all the requirements of the Underwriters' Laboratories, Inc., Publication, Standard for Power-Operated Industrial Trucks.

3.8.1 Air Cleaner: An air Cleaner shall be furnished and mounted underneath of the hood (outside mounting not acceptable). The unit shall be an oil bath type developed and built to filter the air into the intake manifold by passing it over or through a bath of oil and finally through oil wetted mesh to trap the dust particles and other contaminents. The air cleaner shall have sufficient breathing capacity to allow the engine to be operated at wide open throttle without drawing oil from the reservoir. The air cleaner shall be securely connected to the intake manifold inlet and properly supported so that the air cleaner will not become loose or cause damage to the intake manifold. The air cleaner shall be easily accessible when the hood to the engine compartment is opened. The air cleaner shall be constructed to permit easy disassembly for cleaning or removal of mesh.

3.9 Fuel Tank: The capacity of the fuel tank system shall be not less than 85 gallons and constructed so that the tractor engine and the gas turbine compressor may both draw fuel simultaneously. A standpipe shall be provided for turbine suction so that the turbine cannot draw from the bottom 3 gallons of fuel in the tank. The fuel tank shall be mounted in a protected location above tractor frame

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

and shall be equipped with a filler tube, safety filler cap, a float-type rheostat, a fuel shutt-off valve at the fuel line connection, an easily accessible drain plug in the botton of the tank, and an overflow tube which shall be deigned to carry the overflow fuel from engine, muffler, or muffler tail pipe. The tank shall be made of aluminum or stainless steel. The filler tube shall be located in an easily accessible location, of adequate size to receive a 2 inch filling nozzle. The filler tube shall incorporate a removable 40-mesh strainer, and have a chain secured filler cap.

3.9.1 Fuel Lines: Adequate size fuel lines shall be provided to insure optimum operation at all speeds. Fuel lines shall be protected by suitable insulation or heat reflectors in high temperature areas of engine to prevent vapor lock or explosion.

3.10 Instruments: The vehicles shall have installed in the dash panel the following instruments:

Coolant Temperature Gage	Ignition Switch
Torque Converter Temperature Gage	Dash Light ORD 7358618
Oil Pressure Gage	Hour Meter
Light Switches	Ammeter
Starter Switch	Fuel Gage

All instruments and switches shall be back lighted, and have waterproof and dust proof cases.

3.11 Cooling System: The cooling system shall consist of: a radiator with filler cap, water pump, fan and fan belt, thermostat, and connecting hoses and fittings. Two drain outlets shall be provided. One drain cock shall be located in the base of the radiator or in the pump inlet elbow at the lowest point; and the second outlet shall be located at the lowest point of the liquid coolant jacket in the cylinder block. The cooling system shall be of sufficient capacity to prevent coolant temperature exceeding 200 degrees Fahrenheit in an open system, or 225 degrees Fahrenheit in a pressurized close system, as measured at the head outlet when operating continuously under normal operating conditions in an ambient temperature of 125 degrees F. For testing purposes and to avoid the necessity of operation in a 125-degree F ambient temperature, the differential between actual ambient temperature and 125 degrees F will be either added or subtracted from the coolant temperature when operating with the thermostat removed and using the coolant temperature reading taken at the radiator expansion tank. A remote indicating temperature gage, accurate within plus or minus 5 degrees F at 180 degrees F, shall be mounted on the instrument panel. Closed pressure systems shall be provided with a positive operating pressure relief valve that opens at

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the currentversion before use.

less than 7 p.s.i. and a positive operating vacuum breaker valve that opens at a pressure differential of less than 1 p.s.i. Cooling system is to be filled with sufficient anti-freeze to protect the system down to -25° F.

3.12 Radiator: The radiator assembly shall consist of a fin-and-tube or honeycomb type core with an expansion tank at the top and a sediment tank at the bottom. The radiator shall be protected by a heavy front bumper plate or by chassis construction. Where the top of the radiator is lower than the top outlet of the engine, a surge tank may be used.

3.13 Water Pump: The water pump shall be centrifugal type, using a corrosion resistant shaft, if shaft is in contact with water. "Life-time" lubricant system is desirable, but not mandatory. In the event "life-time" system is not supplied, the water pump shall be equipped with accepted commercial type lubricant fittings. No provision for lubrication is required when "life time" bearings are utilized.

3.14 Thermostat: A nominal 180-degree F thermostat shall be located in the cylinder head outlet to the radiator and shall be easily removable.

3.15 Exhaust System: The exhaust system shall consist of: exhaust manifold, a pipe connected to a muffler, and an extended tail pipe. The tail pipe shall not extend beyond the rear end of the tractor, and shall be secured with adequate brackets. The exhaust gases shall be emitted in a horizontal direction at the rear of the tractor. The muffler and exhaust system shall be installed on the tractor so that inadvertent injuries through burns to operating personnel are virtually impossible by proper concealment.

3.16 Muffler: The muffler shall conform to accepted commercial automotive specifications and shall be capable of withstanding repeated backfire emissions (at least 20 effective trials) of the engine without mechanical failure. A spark arresting type muffler is required.

3.17 Transmission: The transmission shall be a heavy duty, automatic planetary, full torque shifting unit equipped with a hydraulic torque converter. The transmission shall have not less than three forward speeds and one reverse speed arranged so that when placed in lowest range it will remain in that range. When placed in "high" range, it shall shift automatically through the lower range or ranges. A "park" position on the controls is not acceptable. Input torque capacity shall be equal to the engine and torque converter maximum output torque. An external oil cooler shall be provided to maintain proper oil

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

temperature under severe and extended operating conditions, and an overheating warning device shall be incorporated. Provisions shall be made to check the transmission oil level from the outside of the tractor. The transmission bands shall be accessible for adjustment. The transmission shall be of the latest type produced by the transmission manufacturer.

3.18 Power Steering: Steering shall be accomplished by a yoke type steering wheel mounted on a steering column. The steering wheel shall turn the front wheels by worm and sector, cam and lever, or approved equal reduction devices which permit easy operation. Clockwise or right hand rotation of the steering wheel shall provide for right hand or clockwise turning of the tractor while in a forward motion. Provisions shall be made for accurate precision adjustment of steering mechanism to compensate for wear of all major steering parts, particularly steering gear reduction mechanisms and for lubrication of all friction points by easily accessible grease fittings or a substantial lubricant reservoir. When a reservoir is provided a readily accessible device for checking lubricant level shall be used. The steering wheel shall be capable of being turned from extreme left or to extreme right by the operator at a force no greater than 10 pounds rim pull with a stationary unloaded tractor on dry brushed or troweled concrete surface.

3.19 Service Brakes: Service brakes shall be either air-type or air-over-hydraulic type.

3.20 Parking Brake: The parking brake shall be an independent brake system mounted on the drive shaft that rotates with the wheels, consisting of either an internal-expanding or an external-contracting type mechanical friction brake. The parking brake shall be capable of holding the tractor stationary on a 30 degree inclined slope. The parking brake shall be independently and mechanically operated by a hand lever equipped with a locking device within easy reach, allowing the safe movement of the operator, both on or off the vehicle, with the brake actuated or released.

3.21 Wheels, Tires, and Tubes

3.21.1 The tractor shall be mounted on six steel or cast iron, disc or spoke-type wheels with non-skid tread pneumatic tires, as specified in paragraph 3.22.3. Each wheel shall be mounted on the type, grade, and class of ball or roller bearings best suited for its function as recommended by the manufacturer. The rear drive wheels shall be dual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

3.21.2 Tire Clearances: When the vehicle is loaded to its gross rated capacity and with the tires inflated to the manufacturer's recommended pressure, sufficient clearance between tires and tractor body shall be provided for the installation of heavy duty tire chains for use in deep snow or on icy surfaces.

3.21.3 Tires: Tires shall be pneumatic, high pressure, conventional, commercial non-directional type tread, conforming to Federal Specification ZZ-T-381. Single tires on front; dual tires on rear of the tractor. Rear tires shall be tubeless. The front tires shall be 6.00 x 9 x 8 ply; rear tires shall be 8.00 x 17.5 x 8 ply.

3.21.4 Tubes: Tubes shall be a heavy duty type conforming to Federal Specification ZZ-I-550 and shall be the proper size to fit the front tires furnished. Valve stems and cores shall be furnished to meet with the tire and rim application so that the tires may be easily inflated.

3.22 Axle Mounting

3.22.1 Front Suspension: The front suspension system shall be such as to prevent any bounce when starting or stopping the tractor under load or no-load conditions.

3.22.2 Rear Axle: The rear axle shall be solidly mounted to the integral frame body and shall be capable of supporting the maximum rated load of the tractor.

3.23 Drive Axle: The drive axle shall be of double reduction gear type. The final reduction shall be in the outer wheel hub. The outer wheel reduction shall be specified. The front wheel spindles shall be tapered and hardened. The axle manufacturer's certificate of approval for the tractor covered by this specification shall be submitted with the bid proposal.

3.24 Integral Frame and Body

3.24.1 Main Frame and Body: The metal of the main frame, cross members, and outer skin shall be of a thickness of 1/2 inch and integrally fabricated from steel plates. The 1/2 inch thickness does not apply to the hood and removable access sections. The hood will conform to the specifications listed in paragraph 3.24.6. Design shall be sufficient to prevent distortion or warping. Structural points shall be adequately welded, and welds shall be thoroughly fused to provide a rigid unit structure. The frame and body structure shall have sufficient depth

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

to afford protection of all working parts. The top of the body shall be capable of supporting a static weight of 600 pounds without damage. A one man cushion seat with a back rest shall be mounted on a spring and foam rubber arrangement located directly behind the steering wheel. The seat shall be fastened by 2 metal hinges constructed of a corrosion resisting material. Due to operating conditions aboard carriers, it is mandatory that all outside corners be rounded as shown in figures 1, 2, and 3.

3.24.2 Rear Counterweight: A detachable rear counterweight shall be furnished with the tractor. It shall be designed to include easily removable tractor rear lights, rear hitch and pintle light, and be equipped with adequate lifting attachments. Accessories and counterweights shall be easily attached to or removed from the tractor with standard tools. The counterweight shall weigh not less than indicated in paragraph 3.26.1.c. All connections for the turbine generator shall be included with the counterweight equipped tractor.

3.24.3 Gas Turbine Compressor and Accessories, Main Frame, and Accessories Enclosure

3.24.3.1 Gas Turbine Compressor Carrier Main Frame: A detachable turbine carrier main frame of the dimensions indicated in Figure 2 shall be furnished. The main frame shall be fabricated of steel plate capable of supporting turbine and all accessories. The frame shall have sufficient strength to absorb the full drawbar pull of the pintle attached to the rear member of the frame. The main frame shall be attached to the tractor by the same arrangement given for the counterweight in paragraph 3.24.2.

3.24.3.2 Turbine and Accessories Enclosure: The enclosure for the gas turbine, 60 KVA, AC generator, DC generator shall be of the design shown in Figure 2. The workmanship and finish on all parts shall be in accordance with high grade aircraft practice for equipment of this type. The enclosure and main frame shall be furnished with four hoisting fittings suitable for hoisting as one unit (the enclosure with gas turbine and accessories installed). Removable cover panels shall be provided on the enclosure to permit access to areas of the compressor and accessories which require routine maintenance and servicing. All doors on the exterior surface of the enclosure shall be attached with full length piano hinges and shall be springloaded to prevent free swinging of the doors. Panels which require frequent removal shall incorporate a flush, quick release, Arctic type latch. Screens shall be provided for the compressor inlet air and turbine exhaust gas outlet so as to produce a minimum effect on the performance of gas turbine compressor mounted within the enclosure.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the currentversion before use.

3.24.3.3 Stowage: Arrangement shall be made for stowage of flexible ducts on the outside of the tractor. This may be accomplished by using clamps or straps. Provisions shall be made for stowing the AC and DC cables on the rear of the enclosure.

3.24.3.4 Lights, Pintle, etc., Quick Attachment: The main frame and enclosure shall be designed to facilitate quick installation and removal, using standard tools, of rear lights, rear hitch and pintle light,and be equipped with adequate lifting attachments. The turbine carrier frame shall be easily installed or removed from the tractor with standard tools.

3.24.3.5 Turbine Operating and Control Panel: The Turbine, AC/DC generators control panel shall be located on the right side of the operator's compartment, and within easy reach of the seated operator. Sufficient space shall be provided from the rear of the turbine control panel to the rear of the tractor to accommodate all instrument and control wiring. Lights shall be provided to illuminate the turbine control panel.

3.24.4 Frame Body Structure, and Turbine Support Test: The main frame body structure and the turbine support structure shall withstand the shock of driving over 2 x 4 inch boards spaced at 4-foot intervals on a 144-foot course. The tractor with the turbine unit carrier,loaded to a weight equal to the turbine unit, and enclosure shall be driven over this course at a speed of 8 miles per hour, 6 times in each direction.

3.24.5 Markings: The entire outside of the tractor body shall be free of all commercial markings, decalcomania, name and date plates except those specified in paragraphs 3.36 and 3.38.

3.24.6 Cowl and Hood: The engine shall be completely enclosed with a one piece hood that is hinged on the front. The hood shall be provided with latches to prevent accidental closure when raised, and accidental opening when vehicle is in motion. The design shall permit removal and replacement of the power unit, by raising (but not removing) the hood. The hood shall be of adequate design to support four 200-pound men without buckling and shall be opened by a lifting force not to exceed 75 pounds. Sides and top shall be rolled from one sheet of metal. No sharp corners are permitted.

3.24.7 Lifting Attachments: The tractor shall be provided with four lifting eyes, properly located to allow lifting of the tractor in a normal position by means of a cable sling without spacer bars. The lifting eyes shall have an inside

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

diameter of not less than 2-1/2 inches, and each lifting eye shall be capable of withstanding a strain of not less than 500 percent of the gross tractor weight at the respective point of attachment. The notation "LIFT HERE" shall be stenciled near each lifting eye.

3.24.8 Tiedowns: Provision shall be made for four tiedown places, two places on the front and two places on the rear of the tractor. These tiedown points shall have a radius of four inches and shall be capable of restraining the tractor, without shifting thereof, while lashed to the deck of the carrier.

3.24.9 Pintle: The pintle shall be similar and equal to Model 400-CA (Holland Hitch Company). One pintle shall be installed on the front and one on the rear of the tractor. Both the counterweight and the turbine carrier shall be drilled and tapped for the rear pintle, also for an 18-inch pintle extension which shall be provided. The extension shall be removable and shall utilize the existing holes in the rear of the tractor. The center line of the front and rear pintle shall be not less than 11 inches above the ground line.

3.25 Tractor Contour

3.25.1 Basic Tractor: The basic tractor, with detachable rear counterweight attached, shall be as outlined in Figure 1.

3.25.2 Tractor Including Turbine Unit: The tractor with the detachable rear counterweight removed, and the turbine unit carriage attached, shall be as outlined in Figure 2.

3.25.3 A general illustration of the required design is shown in Figure 3.

3.25.4 Turbine Carrier Main Frame and Turbine Enclosure: The turbine carrier main frame and turbine enclosure shall be as outlined in Figure 2.

3.26 Minimum Allowable Weights

3.26.1 Basic Tractor:

a. Front end shall weigh not less than 2900 pounds.

b. The rear end with detachable rear counterweight shall weight not less than 8800 pounds.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.

c. A detachable rear counterweight shall be furnished. The detachable rear counterweight shall weigh not less than 2000 pounds and be designed for quick attachment of removable tractor rear lights, rear hitch and pintle light, and be equipped with adequate lifting attachments.

3.26.2 Gas Turbine Compressor and Accessories: When requested, a gas turbine compressor and accessories shall be furnished consisting of the following:

a. Gas turbine compressor

b. Turbine accessories

c. Turbine and accessories enclosure

d. Turbine accessories main frame

3.26.2.1 Gas Turbine Compressor and Accessories Enclosure: The gas turbine compressor and accessories enclosure consists of a sheet metal box enclosing the entire gas turbine and accessories, mounted on a main frame fabricated of steel plate capable of supporting the units encased in the sheet metal box. The frame shall have sufficient strength to absorb the full drawbar pull of the tractor through a pintle attached to the rear member of the frame.

3.27 Hour Meter: The tractor shall be equipped with an electric flush type hour meter which shall register the number of hours of operation of the engine. The hour meter shall be flush-mounted in the instrument panel, and shall be connected in series with the ignition switch and engine lubrication pressure switch. The counting mechanism shall have not less than five numbered wheels, shall count to not less than 9999.9 and shall be the product of a manufacturer regularly engaged in the production of this type instrument.

3.28 Grease Fittings: Grease fittings shall be similar to the types used in standard automotive practice and shall be easily accessible.

3.29 Maintainability: The tractor shall be designed as specified in this document and provide the following features:

a. Minimum number of parts consistent with reliability, operation, and optimum performance

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid, simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance procedures with minimum disturbance to other equipment, parts, or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by accessible location. To aid in performing maintenance activities and if engineering, reasoning, or data determine that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general purpose tools and equipment normally available commercially. The use of special-purpose tools and equipment shall be approved by the Contracting Officer.

e. Require a minimum number of tools for maintenance by improved design practice, such as the reducing of the variety of bolt head sizes to the practical minimum.

3.30 Workmanship: Workmanship shall be of the highest grade throughout and in accordance with the best standard practice for this type equipment.

3.31 Steel Fabrication: Steel used in the fabrication of this equipment shall be free of kinks and sharp bends. The straightening of metal shall be done by methods that will not damage the metal. Shearing and chipping shall be done neatly and accurately. Flame cutting, using a tip suitable for the thickness of metal may be used instead of shearing and sawing. Reentrant cuts shall be made in the best possible manner. All bends of major character shall be made by controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.32 Bolt Connections: Bolt holes shall be accurately punched or drilled and shall have all burrs removed. Rivets shall be driven with pressure tools, and shall completely fill the holes. Rivet heads, when not countersunk or fluted, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

Source: https://assist.dla.mil -- Downloaded: 202640-19T12:54Z
Check the source to verify that this is the currentversion before use.

3.33 Welding: The surfaces of the parts to be welded together shall be free of rust, scale, paint, grease or other foreign matter. Spot, tack or intermittent weld for strength shall not be permitted. Weld penetration shall be sufficient to provide transferences of minimum design stresses through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

3.34 Machine Work: Tolerances and gages for metal fits shall conform to standards of best commercial practice Finished contact and bearing surfaces shall be true and exact.

3.35 Non-Slip Covering: The tractor floor board and top surfaces of tractor shall be covered with a resilient, non-slip floor covering similar or equal to "Scotch-Tred."

3.36 Marking For Identification

3.36.1 Name Plate: The manufacturer shall furnish and affix a corrosion-resistant metal name plate according to MIL-P-514, in an easily visible and safe location on the instrument panel or fire wall of the tractor, with the following information indented or embossed on the plate:

Tractor, Aircraft, Type MD-3

Capacity: 8500 pounds, DBP

Name of Manufacturer: _____

Serial Number: _____

Model Number: _____

Delivery Date: _____ (month and year)

Contract or Order Number: _____

Inspected: _____

Weight: _____

Date of other identification or maintenance manual, whether by tractor serial number or other _____

3.36.2 Instruction Plates: Each unit shall be equipped with instruction plates, suitably located, describing any special or important procedures to be followed in operating or servicing the unit.

Source: https://assist.dla.mil -- Downloaded: 20260-19T12:54Z
Check the source to verify that this is the current version before use.

3.36.3 Instrument Identification: Each instrument shall be identified by an indented or embossed metal plate or on the instrument face. The plate shall be attached to the instrument panel by metal screws.

3.36.4 Identification of Product: Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.37 Dimensions

3.37.1 Detailed Construction: Tractor design shall be suitable to withstand shock loadings Grade A of MIL-S-901 (Navy). The tractor shall conform to the following requirements:

1. Two-wheel drive, two-wheel steer.
2. Drawbar pull 8500 pounds, plus or minus 3 percent.
3. Gross weight of basic tractor, including removable counterweight: 12,000 pounds, plus or minus 3 percent.
4. Maximum weight of removable counterweight: 2400 pounds.
5. Minimum weight on the front wheels: 2900 pounds.
6. Minimum weight on the rear wheels: 8800 pounds.
7. Minimum gross weight, including gas turbine and accessories: 12,000 pounds, plus or minus 3 percent.
8. Maximum height: 36 inches.
9. Maximum length, including removable counterweight, less pintle: 123 inches.
10. Maximum length, including gas turbine and accessories, less pintle: 161 inches.
11. Maximum basic tractor unit width: 65 inches.
12. Maximum turning radius: 124 inches, outside outer tire.
13. Minimum ground clearance: 5 inches.
14. Speed: 15 miles per hour.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

3.38 Finish

3.38.1 Treating and Painting: All exterior surfaces, which are normally painted in commercial practice, shall be thoroughly cleaned and painted with not less than one coat of metallic base primer and two coats of synthetic enamel in accordance with manufacturer's normal practice. The finish color shall be Yellow No. 13538, conforming to Federal Standard No. 595. Gas tank filler spout and cap shall be painted Brilliant Red.

3.38.2 Striping: The rear, flat vertical surface of the vehicle shall be painted Black No. 1775, conforming to Federal Standard No. 595. Then, 4-inch stripes of yellow, reflectorized sheeting shall be applied 4 inches apart, at an incline of 45 degrees from the left and right of vertical to simulate an inverted "V" pattern of yellow and black stripes.

3.38.3 Certification of Compliance with Specification: The bidder shall certify: that the tractor, component units, and parts shall be suitable for the work to be performed; shall be constructed to definite standard dimensions with proper clearance and fits. In addition, the bidder shall submit with the bid proposal, the make, type and model of each component unit, assemblies and accessories, with the manufacturer's certification that it has adequate capacity, and is suitable for the work to be performed by the tow tractor covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection: Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified in this document. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification (where such inspections are deemed necessary) to assure that supplies and services conform to prescribed requirements.

4.2 Examination: Each tractor, including the first production unit, shall be examined for compliance with the requirements listed in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall be examined particularly for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements, or the presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

4.3 Inspection and Tests: The equipment and procedure for preparation for delivery shall be given the necessary inspections and tests as may be necessary to determine conformance with their Specification. The Government reserves the right to use any additional information or procedures deemed necessary to determine compliance with the requirements of this specification.

4.4 Preproduction Samples: The preproduction sample shall be inspected and tested to determine compliance with all provisions of paragraph 4.5 of this specification. The model shall be examined and tested also to determine acceptability on the basis of conformance to accepted engineering design practice and of the efficiency and safety of the operational characteristics normally available in similar commercial equipment.

4.5 Tests on the preproduction test model shall include, but not necessarily be limited to, the following tests:

4.5.1 Preliminary Examination

4.5.1.1 Inspect the tractor for the following conditions:

4.5.1.1.1 Check the coolant level, engine crank case oil, hydraulic oil levels, fuel level, and fuel gage for accuracy at full, one-half full, and empty.

4.5.2 Lubrication Inspection: Observe the lubrication of every grease fitting on the machine in order to determine compliance with the following requirements of the specification:

4.5.2.1 That the tractor is designed for lubrication by standard military lubricants.

4.5.2.2 All grease fittings are easily accessible.

4.5.2.3 Grease fittings in vulnerable positions are protected by recessing or guards.

4.5.3 Check the transmission oil level gage (dipstick) to determine accessibility from outside of body.

4.5.4 Radiated Electromagnetic Interference Test: The tractor shall be tested for radiated electromagnetic interference in accordance with the provisions specified in MIL-STD-462 for REO 5, class 111c, equipment. The radiated interference shall not exceed the limits established by MIL-STD-461 for REO 5, class 111c, items or the tractor shall be considered to have failed the test.

4.6 Tests

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.

4.6.1 Production Sample: Each tractor shall be operated at various speeds, traveling forward and reverse, turning, and stopping to test maneuverability, operation, and general performance in accordance with the requirements of this specification and with the accepted preproduction sample.

4.6.2 Drawbar Pull Tests: On a dry, level, concrete concrete surface, the tractor shall develop a minimum drawbar pull of 8500 pounds, , at a speed of approximately 1 mph. This test shall be conducted not less than four times, and the average drawbar pull shall be not less than 8500 pounds,

4.6.3 Endurance Test: The tractor, with a towed load of 60,000 pounds, shall be driven over a smooth, paved road at not less than 8 miles per hour for a period of not less than 4 hours. During this operation, observe temperature gages for overheating; pressure gages for low pressure; and ammeter for low charge rate. Note anyunusual noises and operating deficiencies. The cooling system shall neet the requirements in paragraph 3.11.

4.6.4 Speed Test: The fully equipped tractor shall be capable of a speed of not less than 15 mph at an engine speed that does not exceed the safe peak engine revolution rate.

4.6.5 Torque Converter Test: Immediately following the test outlined in paragraph 4.6.3, the vehicle torque converter fluid temperature shall be recorded.

With the vehicle transmission in maximum reduction, hold vehicle from moving with the vehicle brakes, and accelerate engine to just below the stall RPM. Hold the vehicle in stationary position for one minute. The converter oil temperature shall not exceed 250°F in an ambient temperature of 125°F. For testing purposes and to avoid the necessity of operation in 125°F ambient, the differential between actual ambient and 125°F will be appropriately added or subtracted.

4.6.6 Braking Test

4.6.6.1 Power Brakes: The power brakes shall be capable of stopping the tractor at least four times from a speed of 8 mph within 10 feet after the engine has been cut off. The tractor shall be capable of coming to a complete stop within 10 feet from the instant of brake pedal application at a speed of 8 mph.

4.6.6.2 Brake "Fade Out" Test: The service brake system of the tractor shall be tested by a brake "fade out" test consisting of bringing the tractor from 15 mph to a full stop 20 times in 30 minutes and then immediately, under maximum acceleration, bringing the tractor to 15 mph and stopping within 80 feet.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the currentversion before use.

4.6.6.3 Parking Brake: The parking brake shall be capable of holding the tractor on a 30-degree grade.

4.6.7 Turning Test: The tractor shall be driven in full 360-degree turns to the right and to the left and the radius of each turn shall be no more than 124 inches.

4.6.8 Pull Test: With a towed load of 40,000 pounds, the tractor transmission shall be shifted through all forward gears without stopping the tractor. On a 10-percent incline, the tractor transmission shall be capable of being shifted to lower gear ranges without stopping and with a minimum effort by the operator.

4.6.9 Alternator Test: The alternator shall be tested to determine if the output complies with paragraph 3.6.3.

4.6.10 Neutral Safety Switch Test: The neutral safety switch shall be tested to determine its compliance with paragraph 3.6.7.

4.6.11 Storage Battery Mounting Check: The battery mounting device shall be checked to determine its compliance with paragraph 3.7.7.

4.6.12 Fuel Tank Check: The fuel tank shall be checked to determine its compliance with paragraph 3.9.

4.6.13 Transmission Test: The transmission shall be tested to determine if its operation complies with paragraph 3.17.

4.6.14 Power Steering Test: The power steering shall be tested to determine its compliance with paragraph 3.18.

4.6.15 Tire Clearance Inspection: The tire clearances shall be checked to determine compliance with paragraph 3.2.1.2.

4.6.16 Axle Mounting Test: The axle mounting shall be inspected and tested to determine its compliance with paragraph 3.22.

4.6.17 Main Frame and Body Test and Inspection: The main frame and body shall be tested and inspected to determine its compliance with paragraph 3.24.4.

4.6.18 Tractor Contour: The tractor shall be inspected to determine its compliance with paragraphs 3.25.1, 3.25.2, 3.25.3 and 3.25.4.

4.6.19 Minimum Allowable Weight: The tractor shall be weighed to determine its compliance with paragraphs 3.26.1 and 3.26.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

4.6.20 Gas Turbine Compressor and Accessories Inspection: The gas turbine accessories enclosure and main frame shall be inspected and tested to determine compliance with paragraphs 3.26.2 and 3.26.2.1, subparagraphs c and d.

4.6.21 The turbine frame and assembly shall be completely installed and removed from the tractors by two men in a maximum time of 2 hours.

NOTE: This shall include removing all lights, pintle, etc. from tractor and re-installing on turbine carriage, and then removing lights, pintle, etc. from turbine carriage and reinstalling on the tractor.

4.7 After Tests: After the foregoing performance tests, and after any individual test made at his discretion, the Government Inspector shall check carefully to determine conformance with the requirements of this specification. Excessive noise in the power train, which,in the opinion of the inspector, is due to defective material or workmanship, shall be cause for the disassembly and repair of the power train of the tractor under test, and of all other tractors of the same contractor where the same fault is revealed.

4.8 Replaced Parts: At the discretion of the inspector, and at the contractor's expense, the part or parts showing failure may be replaced by other parts of sufficient strength and the tractor retested. Failure of any replaced parts or other parts during second test, will be sufficient cause for rejection of the tractor.

4.9 Service Test: The Government reserves the right to conduct an extended service test of 500 miles or more on any tractor being furnished under the contract, as deemed necessary by the Naval Air Systems Command (NASC). Unless otherwise specified, the test will be conducted at a Government proving ground. This test shall be for the purpose of determining mechanical faults, if any, which are due to poor workmanship or the use of inferior materials by the contractor. All such faults revealed by these tests shall be corrected by the contractor, at no expense to the Government, on all tractors where the same faults are revealed. This test may also include the installation of the gas turbine compressor to determine that the tractor meets the requirements aboard ship and of this specification.

In addition to the tests outlined,a manufacturer having no previous experience in building and testing MD-3 tractors shall be required to perform the following service test. The vehicle shall be required to operate with no load over a distance of not less than 5,000 miles over improved roads at a speed of no less than 10 m.p.h. The vehicle shall be required to operate over 2,000 miles under the following listed conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

(1) 100 miles @ 7,500 pounds drawbar pull

(2) 500 miles @ 5,000 pounds drawbar pull

(3) 800 miles @ 4,000 pounds drawbar pull

(4) 400 miles @ 2,500 pounds drawbar pull

(5) 200 miles @ 2,000 pounds drawbar pull

Upon completion of tests, a test report shall be prepared in accordance with MIL-STD-831. Three complete copies of the test report are to be furnished to the procuring activity.

4.10 Production Tests: The Government Inspector shall test at least 4 percent of the units produced under a given contract as outlined under paragraph 4.

5. PREPARATION FOR DELIVERY

5.1 For Domestic Shipment: Unless otherwise specified in the contract or order, the manufacturer's normal commercial practice for domestic shipments will be acceptable.

5.1.1 Additional Processing for Export Shipment to Extra Continental Stations, Below Deck Shipment, and for Immediate Issue:

1. Engine Compartment: Clean and dry engine, temporarily seal all openings in electrical equipment. Spry interior of engine compartment, exterior surfaces, and assemblies, with waterproofing varnish MIL-V-13811, including the following items:

Belt Pulleys	Alternator
Engine Manifold	Voltage Regulator
Exhaust Pipe (in engine compartment)	Wiring Compartment
Fuel Pump	Wiring from Instruments to Fire Wall

Do not spray front or rear of instrument panel. Do not spray ventilation louvres.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the currentversion before use.

2. Engines: After draining fuel system, run engine until fuel is exhausted.

3. Tires: Inflate all tires to 10 p.s.i. above normal operating pressure.

4. Engine: Seal air intake, breather, and crankcase ventilation opening with paper bag and tape.

5. Muffler: Seal outlet with waterpro of material.

5.2 Marking for Shipment: In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use: Tractors covered by this specification are intended for use in towing all carrier based aircraft aboard aircraft carriers and othe aeronautical equipment in all types of weather and in temperatures ranging from minus 25°F to plus 125°F.

6.2 Information to Accompany Bid: Bidder shall submit a positive statement whether or not item bid upon fully conforms to the terms of this specification. In addition, bidder shall furnish complete descriptive literature, flat size drawings conforming to MIL-D-1000, complete technical data covering the equipment he proposes to furnish, and shall identify the equipment by the name of the manufacturer and the model number.

6.2.1 Service Organization: A major factor in approving a manufacturer for the unit described in this specification shall be the ability of said manufacturer to provide an adequate service organization at such major cities as Norfolk, Virginia; Jacksonville, Florida; San Diego, California; and at NAS Alemeda, California; NAS Quonset Point, Rhode Island; NAS Atsugi, Japan. A service organization shall be considered adequate when it can provide:

a. Normal field replacement parts

b. Services of factory trained mechanics

c. Repair and/or overhaul facilities

d. Expedited handling of service problems requiring factory coordination

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

Name and address of bidder's distributor shall be submitted with bid proposal. The contractor shall be required to have available for immediate delivery spare parts for a period of at least five years following delivery of the first unit. Lack of specific and complete information will be sufficient cause for rejection of bid(s). Where bidder's product differs from the specific requirements, each point of difference shall be clearly described. This requirement is set forth to facilitate the review of bids and is not to be construed by the bidder as waiving any of the requirements of this specification.

NOTE: The drawings shall be identical to the item the bidder proposes to furnish and shall show the wheel loads, and center of gravity of the unit.

6.3 Guaranty: Invitation for bids and contracts or orders should specify the following guaranty clause, except in those instances where it is determined that the inclusion of the guaranty clause would not be in the best interest of the Government:

a. "The contractor guarantees that at the time of delivery thereof, the articles provided for under this contract will be free from any defects in material or faulty workmanship or improper design, and will conform to the requirements of this contract. If the Government discovers any such defects or nonconformance within a period of one year after delivery, it may require the contractor: (1) to correct or replace (as designated by the Contracting Officer) the defective or nonconforming articles or part thereof; or (2) to repay such portion of the contract price as is equitable in the circumstances. Notice of the defect or nonconformance shall be given to the contractor within a reasonable time after delivery thereof."

b. "If required by the Government to correct or replace the defective or nonconforming article or part thereof, the contractor shall,with all possible speed, make such correction or replacement. When such correction or replacement requires transportation of the article or part thereof, shipping costs, not exceeding usual charges from site of tractor operation within the continental United States, from the nearest designated port of embarkation to the contractor's plant and return, shall be borne by the contractor; the Government shall bear all other shipping costs. The guaranty shall then continue as to corrected or

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.

replaced articles, or, if only parts of such articles are corrected or replaced, to such corrected or replaced parts; until one year after date of delivery.

Preparing Activity

Navy - AS
Project No. 2420-N021

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

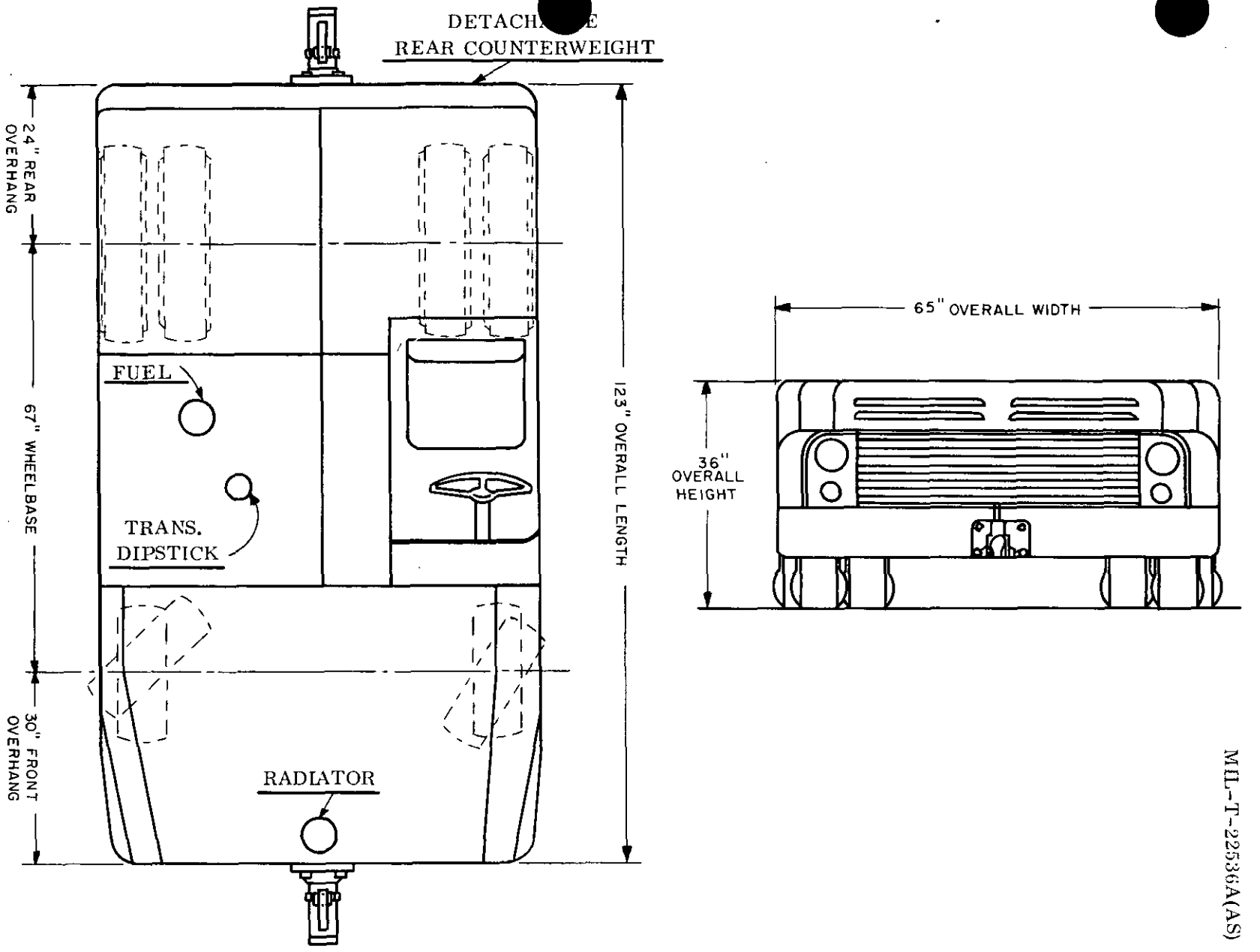


Figure 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

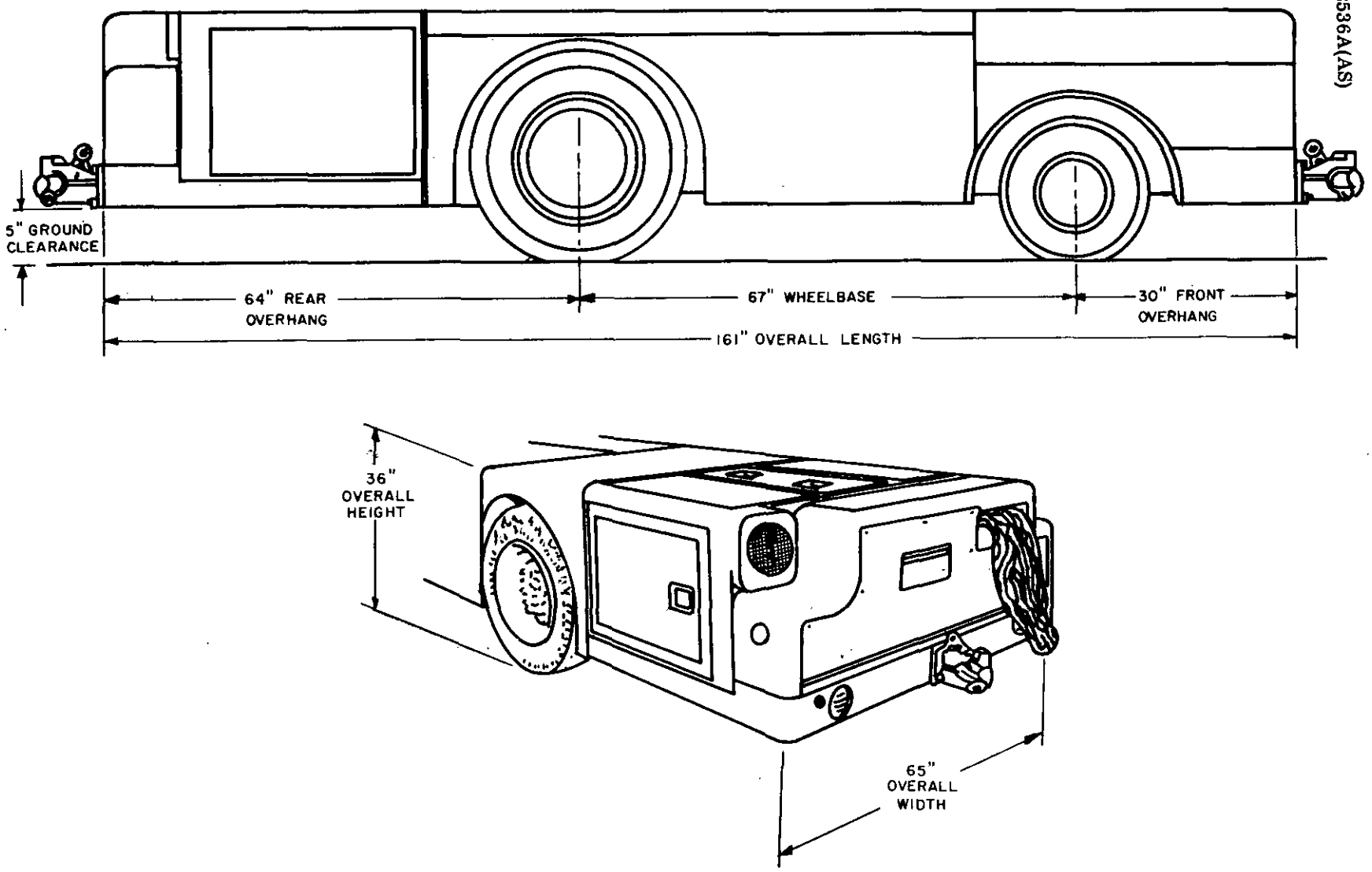


Figure 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:54Z
Check the source to verify that this is the current version before use.

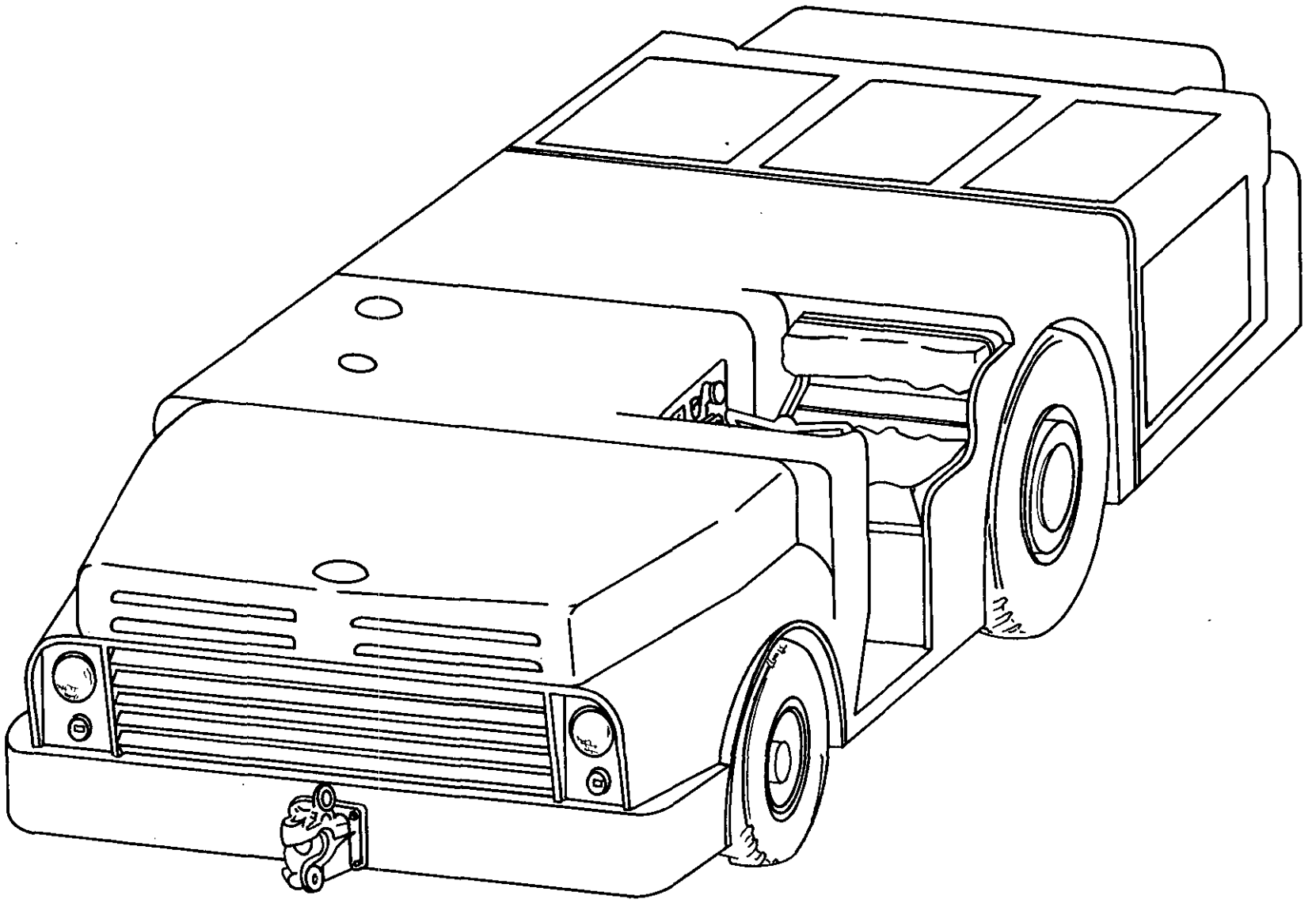


Figure 3

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-22536A(AS) TRACTOR, AIRCRAFT, TOWING, LOW PROFILE SPECIAL DUTY, 8500 LBS. DRAWBAR PULL TYPE MD-3

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8278

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:54Z
Check the source to verify that this is the current version before use.